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One, Two, Three, Four... What's More? Successor Relations, not Mappings to Approximate Magnitudes, Support Learning the Later-Greater Principle

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Abstract

As we count up, number words that occur later in our count denote greater quantities. Here, we tested whether children's understanding of this "later-greater" principle originates from associating number words with approximate magnitudes, or if instead it arises from knowledge of successor relations between individual number words. In Experiment 1 ($N = 60$), we found that children can identify the greater of two small number words (e.g., "six" vs. "ten") based on associative mappings between number words and the ANS but do not rely on such mappings to compare larger numbers (e.g., "twenty-four" vs. "twenty-six"). In Experiment 2 ($N = 60$), we replicate these findings and show that children's knowledge of successor relations between number words predicts later-greater knowledge, while mappings to the ANS do not. We conclude that children's knowledge of successor relations, but not associative mappings to approximate magnitudes, are implicated in learning the later-greater principle.

Keywords: Later-greater; Number words; Approximate magnitudes; ANS; Successor principle

1. Introduction

As adults, we know that 65 million votes is more than 62 million, or that a crowd of 51,000 is more than a crowd of 50,000. We know these things despite the fact that we have never counted this high and despite the fact that these quantities are not discriminable to human perceptual systems. This is because numerate adults understand a global property of counting—

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that numbers which come later in the count list denote greater numerosities—sometimes called the “later-greater” principle. This principle is obvious to numerate adults who understand that larger numbers are composed from smaller ones. For example, in English, unit labels (“one” through “nine”) and decade labels (“twenty” through “ninety”) can be combined to generate numbers between “twenty-one” and “ninety-nine” and to produce even larger numbers when combined with “hundred,” “thousand,” “million,” “billion,” and so forth. However, previous studies suggest that young children lack knowledge of the later-greater principle and often fail to judge that a number like “eight” is more than “six” despite being able to count and label these quantities (Davidson, Eng, & Barner, 2012; Le Corre, 2014). In the present study, we examined this phenomenon and asked how children learn that later numbers represent greater magnitudes.

Although they initially represent counting as a meaningless list of words (Fuson, 2012/1988), at around $2\frac{1}{2}$ years of age, English-speaking children in the United States learn the meaning of “one” and then acquire the meaning of “two” a few months later, learning “three,” and sometimes “four” several months after that in sequence. At these stages, children are considered “subset-knowers” because they know the meanings of only a subset of numbers within their count list (Briars & Siegler, 1984; Frye, Braisby, Lowe, Maroudas, & Nicholls, 1989; Fuson, 2012/1988; Fuson, Secada, & Hall, 1983; Sarnecka & Carey, 2008; Schaeffer, Eggleston, & Scott, 1974; Siegler, 1991; Wynn, 1990). At around $3\frac{1}{2}$ to 4 years of age, children figure out that the counting procedure can be used to accurately construct sets for all larger numbers, and that the last word in a count represents the cardinality of the counted set (Fuson, 2012/1988; Frye, 1989; Schaeffer et al., 1974; Wynn, 1990, 1992). Children at this stage of number word learning are called “cardinal principle knowers” (henceforth, CP-knowers; Gelman et al., 1986; Wynn, 1990, 1992).

On some accounts, the transition to becoming a CP-knower—which involves making an analogical mapping between counting and cardinality—entails knowledge of the later-greater principle. On one view (Carey, 2004, 2009; Sarnecka & Carey, 2008), upon learning the meanings of “one,” “two,” and “three,” the child notices that when they count up from “one” to “two” and then from “two” to “three,” the cardinality of each labeled set also increases by exactly 1 (sometimes referred to as the “successor principle” on analogy to the Peano–Dedekind axioms). A similar view holds that the child may acquire CP by inferring a more global property of counting—the ordinal position of a number word within the count list corresponds to its relative magnitude, such that later numbers are always greater (Spelke & Tsivkin, 2001). Both these accounts predict that CP-knowers therefore must understand the logical relations between number words in the count list.

Contrary to these predictions, however, several studies report that understanding of the later-greater principle may emerge some time after children become CP-knowers (Davidson et al., 2012; Le Corre, 2014; also see Condry & Spelke, 2008; Spaepan et al., 2018). In one study, Le Corre (2014) tested young CP-knowers’ knowledge of the later-greater principle by asking them to make “more” judgments (i.e., to judge, which is more between two number words, m and n). He found that although many children succeeded when comparisons involved at least one small number word (i.e., “two” vs. “three” and “one” vs. “eight”), they often struggled with larger number

words that they otherwise showed evidence of understanding (i.e., “six” vs. “ten” and “eight” vs. “ten”). Le Corre argued that, rather than deploying a principle, children may be relying on associative mappings between number words and non-verbal representations of number to make judgments. On this view, after becoming CP-knowers, children map number words onto non-verbal representations in the approximate number system (ANS), an evolutionarily ancient system that supports our ability to discriminate quantities based on their ratios, in a manner consistent with Weber’s law (e.g., Barth, Kanwisher, & Spelke, 2003; Cantlon et al., 2012; Dehaene, 2011/1997; Gallistel & Gelman, 1992; Nieder, Freedman, & Miller, 2002; Whalen, Gallistel, & Gelman, 1999). Children then use these associative mappings to compare pairs of numbers—for example, to judge that “ten” is more than “six”—without reasoning about the relations between number words themselves (i.e., that “ten” comes later in the count list than “six”). Consistent with this hypothesis, Le Corre found that children who were better at making verbal estimates of dot arrays for quantities between “six” and “ten” (who he classified as “mappers”) were better at making “more” judgments for these numbers, compared to children who did not make reliably different estimates for these numbers (i.e., non-mappers).¹ Based on these data, Le Corre concluded that CP-knowers’ early “more” judgments may be based on mappings to the ANS rather than on a later-greater principle (for similar results with a larger range of numbers, see Davidson et al., 2012).

Although previous studies cast doubt on the idea that learning the cardinal principle involves mastery of the later-greater principle, they leave open when such knowledge arises in development and what might drive it. One possibility, which we call the “ANS induction” hypothesis, is that after becoming CP-knowers children map a small set of number words to magnitudes (e.g., up to “ten”), and gradually notice that as they count to higher numbers, the associated magnitudes consistently increase. Having noticed this, they then make the inductive inference that this is generally true of counting—that later numbers always denote larger approximate magnitudes—thereby allowing them to reason about relations between numbers whose denotations are not perceptually discriminable (e.g., that “twenty-eight” must denote a larger magnitude than “twenty-six” since it comes later in the count list). Another possibility, however, is that the later-greater principle emerges from learning local successor relations between number words, as suggested by Sarnecka and Carey (2008), but that this learning takes place after children become CP-knowers rather than driving the transition to CP. On this “successor induction” hypothesis, as CP-knowers gain expertise in counting, they notice that counting up from “five” to “six” or from “six” to “seven” always involves adding exactly one item for each successive number word. Having noticed this, children might conclude that numbers later in the count list must denote greater quantities, not because they have represented the absolute magnitudes encoded by “six” or “seven” but instead because they know that counting up entails adding items (e.g., “twenty-eight” must be more than “twenty-six” because it requires adding additional objects). Consistent with this account, previous studies suggest that such knowledge of successor relations—that is, that counting up 1 word in the count list corresponds to adding 1 item—emerges gradually after children become CP-knowers (Cheung, Rubenson, & Barner, 2017; Chu, Cheung, Schneider, Sullivan, & Barner, 2020; Davidson et al., 2012; Schneider et al., 2020; Schneider, Pankonin, Schachner, &

Barner, 2021; Schneider, Sullivan, Guo, & Barner, 2021; Sella & Lucangeli, 2020; also see Fuson, 2012/1988; Secada et al., 1983), and is absent in subset knowers (Schneider, Pankonin, Schachner, & Barner, 2021; cf., Wege et al., 2025; for discussion, see Sella et al., 2021).

2. The present study

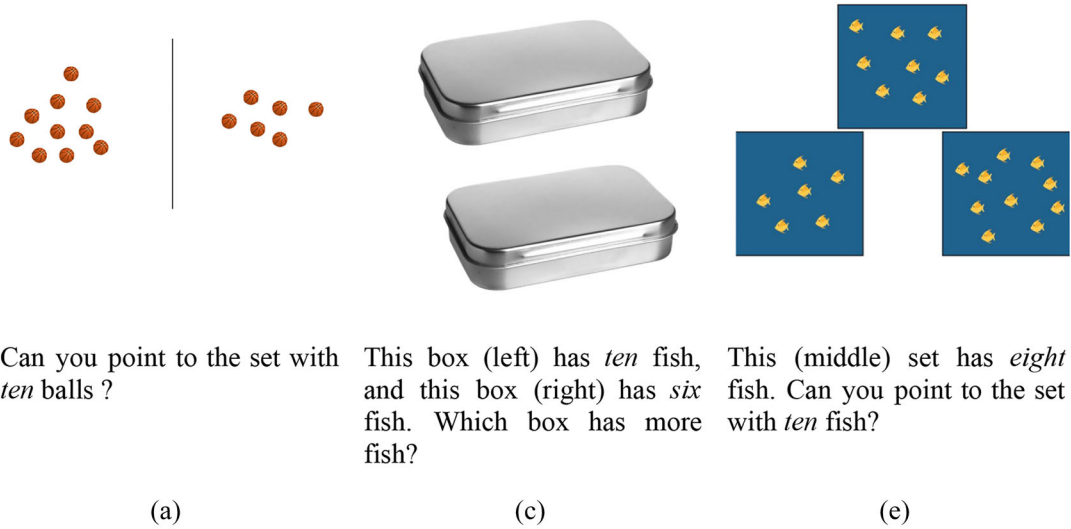
In the present study, we sought to test when and how children learn a generalized later-greater principle while contrasting the relative roles of associative mapping and knowledge of successor relations in this developmental change. To do so, we asked two distinct, but related questions. First, in Experiment 1 we asked whether individual children's "more" judgments were based on associative mappings to the ANS (as in Le Corre, 2014), or if they could also make "more" judgments for pairs of numbers for which they either lack mappings or that are not discriminable using the ANS, consistent with having knowledge of a later-greater principle. Second, given evidence that some children have knowledge of a principle in Experiment 1, we asked what sources of learning might inform their knowledge in Experiment 2. In particular, we asked whether children's later-greater knowledge is better predicted by mappings to the ANS (the ANS induction hypothesis) or by knowledge of +1 successor relations between number words within their count list (the successor induction hypothesis).

To test knowledge of a later-greater principle in Experiment 1, we reasoned that a child who deploys a generalized later-greater principle, like adults, should be able to judge that a large number like 28 is more than 26 even if they cannot discriminate these quantities perceptually. By contrast, a child who relies on associative mappings to the ANS and who lacks a generalized principle should fail for such quantities so long as they cannot reliably differentiate them perceptually since ANS-based discrimination is constrained by Weber's limit (e.g., Barth et al., 2003; Cantlon et al., 2012; Dehaene, 2011/1997; Gallistel & Gelman, 1992; Halberda & Feigenson, 2008; Halberda, Mazocco, & Feigenson, 2008; Nieder et al., 2002; Whalen et al., 1999). Also, children who rely only on associative mappings should fail to make "more" judgments for pairs of numbers that are not mapped to the ANS. To test this, we classified children into "mappers" and "non-mappers" for individual numbers (Davidson et al., 2012; Le Corre, 2014) to assess whether mappings were required for "more" judgments, or if children could make them despite lacking mappings, as predicted by a generalized later-greater principle.

In Experiment 2, we tested the mechanisms that might drive acquisition of this principle. First, to test the ANS induction hypothesis, we asked whether individual differences in ANS mappings predicted children's later-greater judgments for large and perceptually indistinguishable quantities. Second, to test the successor induction hypothesis, we asked whether children's knowledge of the later-greater principle was predicted by their understanding of successor relations between individual number words.

To explore these ideas, we tested children on four tasks. First, to assess children's mappings between number words and the ANS, we used the Point to X Task (Wynn, 1992; see Fig. 1a,b). In this task, children are asked to assign a number word to one of two non-symbolic arrays—for example, "Here are two sets of fish. Can you point to the set with X fish?" Suc-

Small Number Comparisons



Large Number Comparisons

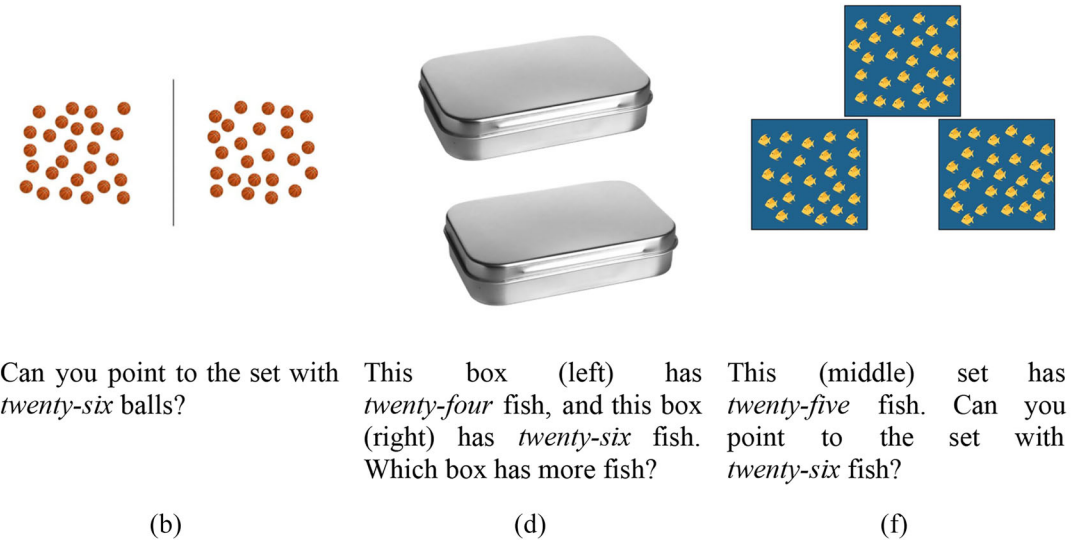


Fig. 1. Examples of numerical comparisons in the Point to X task for (a) small and (b) large numbers; Invisible Later-Greater for (c) small and (d) large numbers (graphic represents physical tin boxes used), and Visible Later-Greater for (e) small and (f) large numbers.

cess on the Point to X task requires the ability to discriminate the two visible sets, as well as mappings between the two magnitudes and the numbers that represent them. For example, if the child is presented with a 5 versus 10 comparison, they should only succeed if they are able to perceptually discriminate the two sets, and also if they have associated the words “five” and “ten” with their respective magnitudes. This task does not require knowledge of the later-greater principle but allows the identification of large number comparisons in which ANS could not support performance. Further details of how associative mappings were calculated are described in the Methods.

Second, in both Experiments, we measured understanding of the later-greater principle using the “Invisible Later-Greater” Task (referred to as the “More” task in Davidson et al., 2012; Le Corre, 2014; see Fig. 1c,d). In this task, children were presented with two boxes and told, for example, “This box has *six* fish and this box has *ten* fish. Which box has more fish?” For numbers that are discriminable via the ANS, like 6 versus 10, mappings to magnitudes might drive judgments. However, for comparisons like 26 versus 28, mappings to the ANS should not explain judgments (unless children can perceptually discriminate them), making it more likely that they have a later-greater principle. For this reason, the Invisible Later-Greater task provides a direct measure of the later-greater principle when used to test quantities that are non-discriminable.

Third, in both experiments, we presented children with a novel, “Visible Later-Greater” task (Fig. 1c,d), in which they were asked to assign a label to one of three visual arrays, when the label for the middle-sized set was already given. For example, children were presented with three sets of objects (e.g., 4, 6, and 8 fish) and were told, e.g., “Here are three sets of fish. This group has *six* fish. Can you point to the set with *eight* fish?” Crucially, to successfully identify the set with “eight” fish, the child not only needed to know that the label “eight” denotes a greater quantity than “six” but also that the set containing 8 items is more than the set with 4 and 6, and therefore must be the correct response (i.e., they must know the ordinal relations between the words but also be able to discriminate their referents). The benefit of this task is that it incorporates the logic of the first two tasks into a single method since success depends upon both knowledge of the later-greater principle (as in the Invisible Later-Greater task) and also the ability to discriminate magnitudes (as in the Point to X task). This is important because in the scenario where children succeed on Invisible Later-Greater for a large comparison like 24 versus 26, it remains possible that this is driven by ANS mappings (though in general this comparison should be indiscriminable). A success for this comparison in the Visible Later-Greater task would support the conclusion that children use the ANS, whereas a failure would suggest that they must use a principle instead.

Finally, in Experiment 2, we test children’s knowledge of successor relations between individual numbers using the Unit Task (Sarnecka & Carey, 2008). In this task, children are shown a container that they are told contains n number of items (“This cup has n objects”), and then see 1 object added to the cup, and are asked, “Are there $n + 1$ or $n + 2$ in the cup?” We asked whether performance on this test of successor knowledge predicted children’s understanding of the later-greater principle, either independently of, or in conjunction with mappings to the ANS.

3. Experiment 1

In Experiment 1, we asked whether children have a principle at all or instead rely on item-based associative mappings to the ANS to make “more” judgments. We reasoned that children who have a generalized later-greater principle should perform differently on the Invisible and Visible Later-Greater tasks. In particular, children should succeed on comparisons of small numbers (e.g., 6 vs. 10) on both tasks since these can be solved by either a later-greater principle or by associative mappings (recall that previous studies find that many CP-knowers have mappings between such small number words and the ANS, and can make comparisons based on these mappings; Davidson et al., 2012; Le Corre, 2014; Le Corre & Carey, 2007). However, for comparisons of large numbers (e.g., 24 vs. 26), children’s judgments may look different on each task depending on whether they use associative mappings or a later-greater principle. If children appeal only to associative mappings, they should fail to compare large numbers with tight ratios numbers on both the Visible and Invisible Later-Greater tasks since such ratios should be non-discriminable via the ANS. Conversely, if children rely on a later-greater principle to make judgments, they should succeed on the Invisible task even if they fail on the Visible task since their knowledge does not depend on the perceptual discriminability of sets.

Second, for children who showed evidence of a principle, we asked if such knowledge differed based on children’s knowledge of mappings between magnitudes and number words. To do this, we used Point to X performance to classify children into “mappers” and “non-mappers” (following Davidson et al., 2012; Le Corre, 2014; Le Corre & Carey, 2007), and compared their performance on corresponding Invisible Later-Greater trials. If knowledge of mappings plays a role in learning the later-greater principle, then mappers and non-mappers should differ in their performance on the Invisible Later-Greater task. However, if mappings to the ANS do not play a critical role, mappers and non-mappers may not differ significantly.

3.1. Method

All methods and analyses were pre-registered on the Open Science Framework (OSF), unless otherwise stated. Pre-registrations and data for both experiments are available via <https://osf.io/nwyqe/overview>.

3.1.1. Participants

Participants were 60 (31 female) typically developing, English-speaking children between 3;2 and 5;11 years of age ($M = 58.41$ months; $SD = 7.88$ months; range = 38–71 months) recruited through a local museum in San Diego, United States. One additional participant was excluded for not completing all tasks.² We did not collect data related to children’s race/ethnicity or socioeconomic status. County data suggest that children aged 3–10 years in San Diego County are mostly Hispanic/Latino (43.3%) and White (37.0%). Other groups include African American/Black (4.8%), Native American (0.4%), Asian (8.9%), Hawaiian/Pacific Islander (0.4%), and multiracial (5.1%) children, as per the KidsData Project

(2021). The median annual household income in the region is \$102,285 (US Census Bureau, 2024).

3.1.2. *Materials and procedure*

All participants were first given two screening measures: (1) Highest Count and (2) Give-a-Number. If children passed the screening (based on criteria described below), they were included in the study and completed the three numerical comparison tasks. The Invisible Later-Greater task was administered in three trial blocks interspersed between the Point to X and Visible Later-Greater tasks. The Point to X task was always administered first, and the Visible Later-Greater task was administered last.

3.1.2.1. *Highest Count:* This task served as a proxy for exposure to number words. Children were asked to count as high as they could with the following prompt, “Can you start counting as high as you can?” If children did not begin counting, they were prompted with “Let’s start together — one...” with rising intonation. The last number reached before stopping or making an error was recorded as the child’s highest count. Children who did not make errors or stop themselves were stopped by the experimenter upon reaching 100. Children with a highest count below “ten” were not included in the study because exposure to numbers up to 10 was needed to complete all subsequent tasks.

3.1.2.2. *Give-a-Number:* An abbreviated version of the Give-a-Number Task (Wynn, 1992) was used to screen children for their CP knowledge. A bowl of multiple objects of the same kind (blue plastic fish) was placed in front of the child. The experimenter said, “Here are some fish and here is a plate. Can you put N fish on the plate?” If the child gave N fish on a request for N , the experimenter asked for $N + 1$ fish on the next trial. If the child failed to give N fish on a request of N , the experimenter prompted them to count to verify that they provided N (i.e., “Is that N fish? Can you count and make sure?”). This follow-up was included to give children every opportunity to succeed on Give-a-Number and demonstrate their understanding of the counting procedure. Because our goal was to identify CP-knowers, trials began with a request for “five” fish, and the highest number requested was “six.” If children succeeded at giving “six” fish twice, they were classified as CP-knowers and included in the study. A maximum of 4 trials (5, 6, 6, 6) were administered.

We used this abbreviated version of the Give-a-Number task because although various versions of the task exist in the literature, and are often administered in different ways (see for discussion: Krajcsi, 2021; Marchand, Lovelett, Kendro, & Barner, 2022), children who are able to count and generate sets of six are generally classified as CP-knowers in past studies of the later-greater principle (Davidson et al., 2012; Le Corre, 2014) as well as in many other studies on numerical development (e.g., Ansari et al., 2003; Halberda et al., 2008; Le Corre, Van de Walle, Brannon, & Carey, 2006; Sarnecka & Carey, 2008; Schneider et al., 2020; Schneider, Sullivan, et al., 2021; Wynn, 1990, 1992), making this a stable basis for comparison of studies across the literature.³

3.1.2.3. Point to X: Children were asked to assign a label to one of two visual arrays—for example, “Here are two sets of fish. Can you point to the set with *ten* fish?” Four comparisons were queried: 6 versus 10, 8 versus 10, 24 versus 26, and 26 versus 30. Each comparison was presented a total of six times. Children were asked to point to the array labeled with the smaller number word on half of the trials and the larger number word on the other half of trials in alternate blocks. Trial order was pseudo-randomized within each block. See Fig. 1a,b.

3.1.2.4. Invisible Later-Greater: The experimenter introduced the child to a stuffed toy animal, “This is Mister Monkey. Mister Monkey is very hungry and loves to eat fish.” The experimenter then showed the child two open tin boxes, each of which contained small plastic fish and said, “This box has *one* fish and this box has *two* fish. Mister Monkey wants this box because two fish is more than one fish!” Practice trials were administered in the following order: 1 versus 2, 2 versus 4, and 1 versus 2. The order in which the larger number word was presented and the side of the table on which it was placed (either left or right) was also pseudo-randomized. For test trials, the experimenter closed the tin boxes so that children based their choices on their understanding of the meaning of the queried number words (“This box has *m* fish and this box has *n* fish. Which box has more fish?”). Queried comparisons were the same as those used in the Point to X task. Each comparison was presented three times making a total of 12 trials. Each queried comparison was presented once in each block, and order of presentation was pseudo-randomized within each block. Before starting each trial block, children were given an attention check, “Look! Mister Monkey is back! You remember what Mister Monkey likes eating? That’s right! Mister Monkey likes eating fish” and reminded that the stuffed animal was hungry and wanted the box with more fish (“Mister Monkey is hungry again and wants the box with more fish”). See Fig. 1c,d.

3.1.2.5. Visible Later-Greater: Children were asked to assign a label to one of three arrays, when the label for the middle-sized set was already given, and the remaining two numbers were pairs used in the Point to X and Invisible Later-Greater tasks. For example, children were shown three sets of objects (e.g., 4-6-8 fish) and told, “Here are three sets of fish. This set has *six* fish. Can you point to the set with *eight* fish?” Children were queried on the same comparisons as the Point to X and Invisible Later-Greater tasks, such that the arrays had the following configurations: 6-8-10, 8-9-10, 24-25-26, and 26-28-30. Each trial was presented thrice for a total of 12 trials. See Fig. 1e,f.

3.2. Results

Our primary analyses were based on a pre-registered plan available on the OSF, though the findings reported below also include several exploratory and post hoc tests. All analyses were conducted in R studio 4.5.1., primarily using the lme4 (Bates & Mächler, 2015), emmeans *emmeans* (Lenth, 2025), and tidyverse (Wickham et al., 2019) packages. All 60 children included in the experiment qualified as CP-knowers (determined by the abbreviated Give-a-Number task) and had highest counts of at least 10. Among those included in the

study, highest counts ranged from 10 to 100 ($M = 42.2$; $SD = 34.20$; $Mdn = 28.5$; Supporting Information Fig. S1).

For all analyses, we began with a base generalized logistic mixed effects model and iteratively added terms for additional variables to the base model. To test for the significance of specific independent variables, we conducted likelihood ratio tests comparing models with and without particular effects of interest. All models contained a random effect of participant, expressed as (1|PID). Best-fitting models were chosen based on reduced AIC value and a significant chi-squared statistic. Analyses are provided in tables within the text or in the Supporting Information. All subsequent pairwise comparisons were adjusted for multiple comparisons using a Bonferroni correction.

In our first set of analyses, we compared children's performance on the three numerical comparison tasks. Recall that the three comparison tasks were designed to assess children's reliance on later-greater principle knowledge or associative mappings to the ANS when making "more" judgments. We reasoned that if children use a generalized later-greater principle as the basis for their judgments, then they should succeed on easier ratios for all tasks, but for tight-ratio (i.e., large number) trials only on the Invisible Later-Greater task since both the Point to X and Visible Later-Greater tasks require the ability to visually discriminate sets via the ANS. To test this prediction, we first examined baseline differences in children's accuracy on the three tasks by constructing a base generalized logistic mixed model with the following formula: Accuracy (0/1) $\sim$ Task (Point to X/Invisible Later-Greater/Visible Later-Greater) + (1|PID). Children's overall accuracy on the Invisible Later-Greater task did not differ significantly from accuracy on the Point to X ($B = -0.006$, n.s.), but was significantly better than on the Visible Later-Greater task ($B = -0.23$, $p < .04$). We examined differences in children's accuracy across small and large trials by adding a set size (small/large) $\times$ task (Point to X/Invisible Later-Greater/Visible Later-Greater) interaction term to the base model, which significantly improved model fit ($\chi^2(1) = 124.66$, $p < .001$), suggesting that children's numerical judgments across tasks varied as a function of the magnitudes being compared. We explore these differences below. Finally, adding fixed terms for children's age ($\chi^2(1) = 6.43$, $p = .01$) and their highest count ($\chi^2(1) = 6.60$, $p = .01$) further improved model fit, but a maximal model that included both did not significantly improve model fit relative to one that included age alone, suggesting that participants' highest count did not explain unique variability in children's judgments (Table 1).

To understand these results, we conducted planned pairwise comparisons examining differences between tasks. We first report differences in small number trials (i.e., 6 vs. 10 and 8 vs. 10), and then large number trials (i.e., 24 vs. 26 and 26 vs. 30). First, for the 6 versus 10 trial, children were more accurate on the Invisible Later-Greater task relative to the Point to X task ($B = -1.37$, $p < .0001$) and the Visible Later-Greater task ($B = -0.89$, $p < .001$), but there were no differences in children's accuracy on the 8 versus 10 trial (Invisible Later-Greater vs. Point to X [$B = -0.31$, n.s.]; Invisible Later-Greater vs. Visible Later-Greater ($B = 0.40$, n.s.)). This suggests that for these relatively small numbers, some children made later-greater judgments for numbers that they could not perceptually discriminate—for example, that they could judge that "ten" is more than "six" but not identify which of two sets of 10 and 6 dots corresponded to which number and was more. Next, we turned

Table 1
Regression models predicting children’s accuracy on the three numerical comparison tasks in Experiment 1

#	Model	Coefficient Estimates (B)				Summary Statistics		
		Age	HC	Trial Size	VLG	PTX	Trial Size *PTX	Trial Size *VLG
1	Task				−0.23*	−0.006		
2	Task * Trial Size			0.16	−0.61***	−0.62***	1.43***	0.81***
3	Age + Task *	0.21**		0.16	−0.61***	−0.62***	1.43***	0.81***
4	Trial Size		0.22**	0.16	−0.61***	−0.62***	1.43***	0.81***
4	Age + HC + Task * Trial Size	0.14	0.15		−0.61***	−0.62***	1.43***	0.81***

Note. Reference Group—Task: Invisible Later-Greater. Trial Size: Large. Note: A random effect of participant (I|PID) was included in all models
Abbreviations: HC, Highest Count; PTX, Point to X; VLG, Visible Later-Greater.
Significance codes: *** $p < .001$; ** $p < .01$; * $p < .05$.

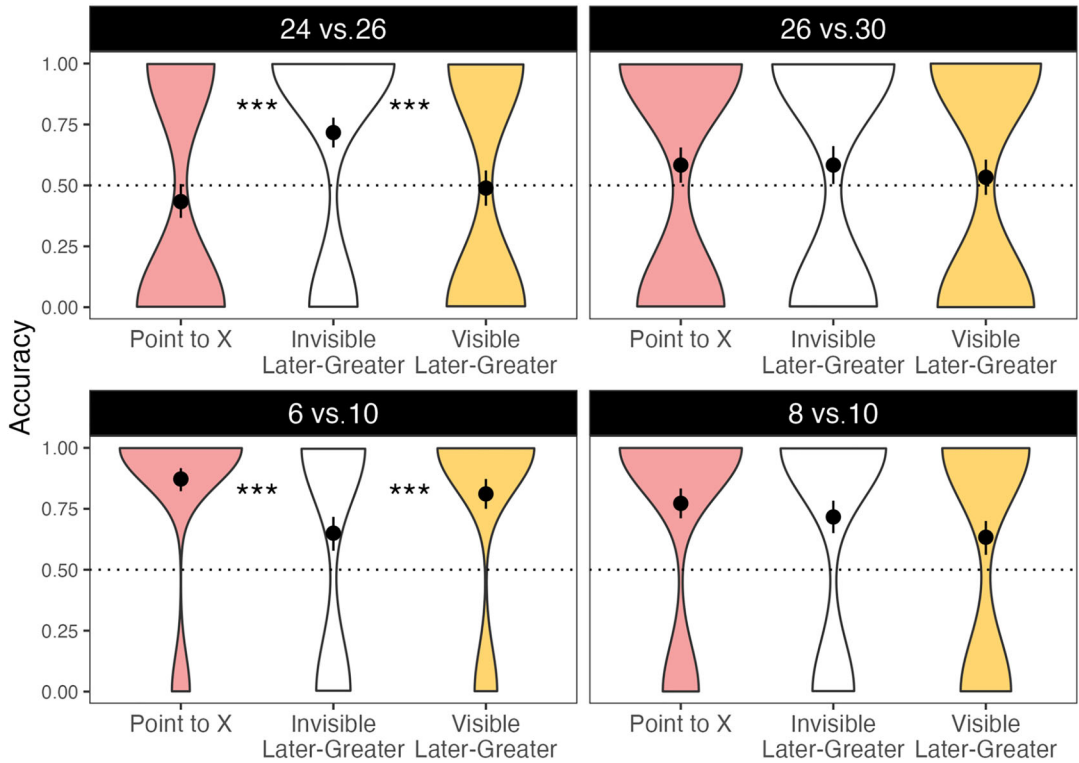


Fig. 2. Mean accuracy on the numerical judgment tasks by color. Each facet represents the queried comparisons, and violins represent the distribution of responses for each task on that trial. The horizontal dotted line represents chance. Error bars represent 95% bootstrapped confidence intervals. Asterisks represent significant differences between tasks. Plots were generated using ggplot2 (Wickham, 2016).

to larger number comparisons, which were the “critical” trials. Here, we found that children were significantly more accurate in making “more” judgments on the 24 versus 26 trial in the Invisible Later-Greater task relative to both the Point to X ($B = 1.28, p < .001$) and Visible Later-Greater tasks ($B = 1.04, p < .001$), compatible with knowledge of the later-greater principle. Somewhat surprisingly, however, on the 26 versus 30 trial, children did not show a similar difference in performance across tasks (Invisible LG vs. Point to X [$B = -0.02, n.s.$]; Invisible LG vs. Visible LG ($B = 0.19, n.s.$)), and performance on this trial was generally poor (Fig. 2). This might be because children do not have associative mappings for these number words but might also be explained by the fact that “thirty” is a decade term, and was therefore either less familiar to them, or did not contain unit labels (from “one” through “nine”) to use as a basis for judgments.⁴ We return to this idea in the General Discussion.

We next conducted t -tests against chance to examine children’s performance on the large number trials of the two Later-Greater tasks, once again reasoning that children using a principle should perform above chance levels on the Invisible Later-Greater task, even if performance does not differ from chance on the Visible Later-Greater task. Consistent with this

prediction, children were above chance on 24 versus 26 ($t = 6.43, p < .001$) and 26 versus 30 ($t = 2.26, p < .02$) on Invisible Later-Greater. However, we did not see similar performance on the Visible Later-Greater task for 24 versus 26 ($t = -0.29, n.s.$) or 26 versus 30 ($t = 0.89, n.s.$). Finally, children's Point to X performance was above chance on 26 versus 30 ($t = 0.89, p = .02$) but not 24 versus 26 ($t = -1.79, n.s.$). Taken together with the pairwise comparisons described above, these data suggest that some "more" judgments are made using a later-greater principle and not on the basis of mappings to the ANS.

To further explore the role of the ANS in children's performance, in a second set of analyses we adopted a strategy similar to Le Corre (2014), which examined whether "mappers" performed similarly to "non-mappers" on the Invisible Later-Greater task (since mappings to the ANS are required for comparisons on the Visible Later-Greater task). In previous studies (e.g., Davidson et al., 2012; Le Corre, 2014; Le Corre & Carey, 2007), children were considered mappers and non-mappers based on the slope of their performance on an estimation task and the assumption that a child who has mappings for a number like "ten" should also have mappings for all smaller numbers. Here, we took a different approach and categorized children as "mappers" and "non-mappers" for each trial based on their Point to X performance. If children got 4 out of 6 trials correct on the Point to X trial for a given comparison, they were considered a "mapper" on that trial.⁵ In this way, a child categorized as a mapper on one trial (e.g., 6 vs. 10) might not be a mapper on another trial (e.g., 24 vs. 26), which we reasoned should provide a more precise characterization of whether mappings to the ANS are involved in specific judgments (i.e., relative to a classification based on the slope of estimates). In pairwise comparisons for both small and large number trials, we found no significant differences between mappers and non-mappers on the Invisible Later-Greater task (Fig. 3; Supporting Information Table S1). This suggests that differences in the ability to map numbers to the ANS did not explain children's "more" judgments, contrary to Le Corre (2014) and Davison et al. (2012), and consistent with the use of an alternative mechanism, like the later-greater principle.

3.3. Discussion

In Experiment 1, we tested whether children rely on associative mappings between number words and the ANS to make "more" judgments, or, if instead, they reason about the relative magnitude of number words based on their ordinal position in the count list—that is, by applying a later-greater principle. We found two main results. First, we found that for the 24 versus 26 trial, children made more accurate "more" judgments on the Invisible Later-Greater task relative to the Point to X and Visible Later-Greater tasks (though when the decade term "thirty" was involved, they performed poorly). This suggests that children made "more" judgments without relying on mappings to the ANS, compatible with later-greater knowledge. Second, we found no evidence that children who had mapped number words onto approximate magnitudes ("mappers") differed from those who had not ("non-mappers") on the Invisible Later-Greater task, again suggesting that mappings to the ANS may not explain success on large number comparisons.

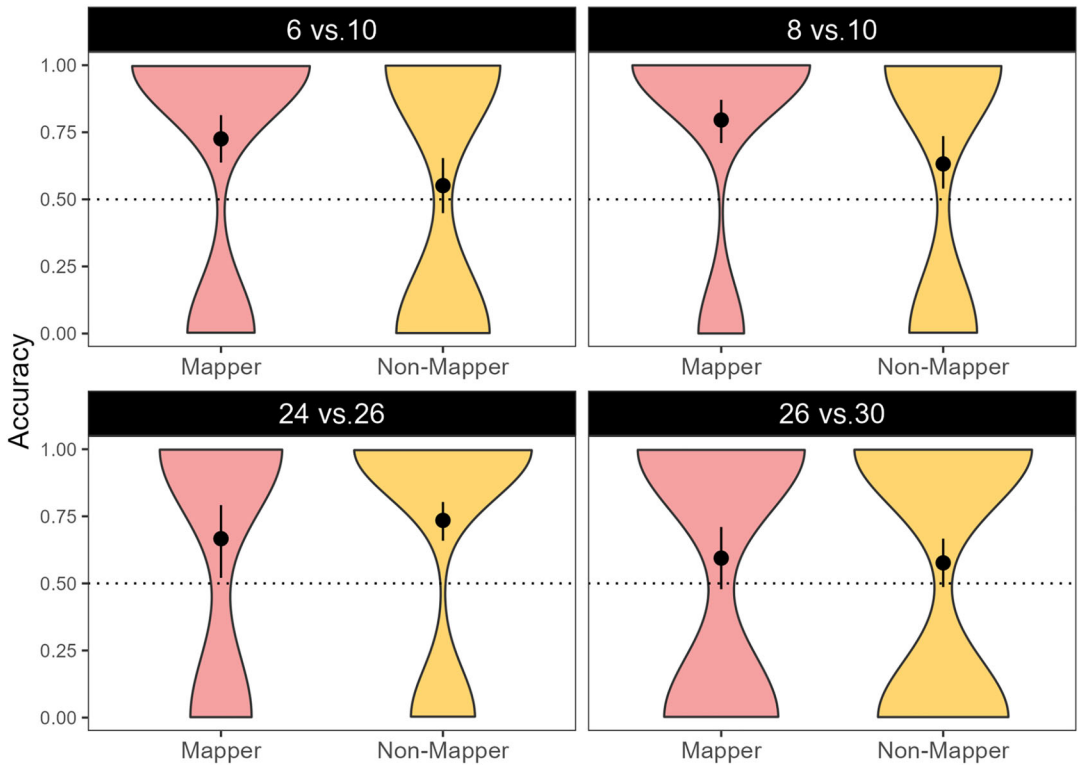


Fig. 3. Mean accuracy on Invisible Later-Greater judgments across mappers and non-mappers. Each facet represents the queried comparisons, and violins represent the distribution of responses for each group on that trial. The horizontal dotted line represents chance. Error bars represent 95% bootstrapped confidence intervals. No significant differences were found between mappers and non-mappers.

Given this preliminary evidence that some children have later-greater knowledge, how might such knowledge arise? One possibility, noted in the Introduction, is that children may learn the later-greater principle on the basis of mappings to magnitudes, by noticing that as we count up, the magnitudes associated with number words increase (ANS induction). Given sufficient evidence, children might make the inductive inference that this property is true of all number words—not just those with mappings to the ANS—and that counting up always corresponds to increased magnitude. Alternatively, it is possible that children acquire the later-greater principle based on knowledge of ordinal “successor” relations that exist between the number words themselves (successor induction). On this view, children might learn later-greater relations by noticing that counting up one number word in the count list always corresponds to adding one object—that the successor of any number n has a cardinal value of $n + 1$, and therefore denotes a greater number. From here, they might generalize this observation about relations between particular numbers (and their precise differences) to all numbers and that in general later number words must always denote greater quantities.

In Experiment 2, we sought to replicate the findings of Experiment 1, while also addressing the mechanisms that might drive acquisition of the later-greater principle. In particular, we asked whether knowledge of the later-greater principle is better predicted by children's prior mastery of successor relations or by mappings of number words to the ANS.

4. Experiment 2

In Experiment 2, we had two goals. First, we tested whether the results of Experiment 1 were replicable. To do so, we compared children's performance on the Invisible and Visible Later-Greater tasks on three large number trials that each featured a difference of exactly 2: (1) 24 versus 26, as in Experiment 1, (2) 26 versus 28, and (3) 28 versus 30. Second, for children who showed evidence of a principle, we asked if such knowledge was more strongly predicted by knowledge of successor relations between individual numbers or to knowledge of mappings between number words and the ANS.

To evaluate the ANS induction hypothesis, we asked whether children's knowledge of mappings to ANS (measured by the Point to X task) predicted performance on the Invisible Later-Greater task. To assess the successor induction hypothesis, we asked whether performance on the Invisible Later-Greater task was predicted by knowledge of successor relations between number words (rather than mappings to the ANS). To do so, we tested children with the Unit Task (Davidson et al., 2012; Sarnecka & Carey, 2008), in which children are told that a container holds a certain number of objects—for example, “five” fish—they watch as 1 or 2 objects added, and they are then asked if there are, for example, “six” or “seven” fish in the container. Crucially, this task requires children to represent successor relations between number words—for example, that “six” comes immediately after “five” and contains one more item—and to express this knowledge outside of the context of a counting routine. We reasoned that only children with strong knowledge of successor relations should perform well on this task, and it might support the inference that as one counts higher, numbers grow larger, compatible with the later-greater principle.

4.1. Method

All methods and analyses were preregistered on the OSF, unless otherwise stated.

4.1.1. Participants

Participants were 60 (22 female) typically developing, English-speaking children between 3;8 and 6;8 years of age ($M = 58.05$ months; $SD = 9.10$ months; range = 44–80 months) recruited from local daycares, preschools, and museums in San Diego County. Two additional children were excluded for not completing all tasks.

4.1.2. Measures and procedure

The procedure remained the same as in Experiment 1, with two exceptions: (1) the trials queried in each task were changed, as described below, and (2) the Unit Task (Sarnecka & Carey, 2008) was administered at the end of the task battery.

4.1.2.1. Point to X: Children were queried on three small number comparisons: 4 versus 6, 6 versus 8, 8 versus 10. This task was used to measure whether children associated small number words to the ANS. The task was administered in the same way as Experiment 1.

4.1.2.2. Invisible Later-Greater: Children were probed on three large number comparisons: 24 versus 26, 26 versus 28, and 28 versus 30. The decision to use 28 in the last comparison (instead of 26 as in Experiment 1) was made to keep the same distance between number words being compared on each trial (N vs. $N + 2$). The task was administered in the same way as Experiment 1.

4.1.2.3. Visible Later-Greater: Children were tested on the same comparisons as the Invisible Later-Greater task. Consequently, the visual arrays had the following configurations: 24-25-26, 26-27-28, and 28-29-30. The task was administered in the same way as Experiment 1.

4.1.2.4. Unit Task: The Unit Task followed the same structure as Sarnecka and Carey (2008). The experimenter showed children a cup with N objects (e.g., lemons), saying, “Look! There are N lemons in a cup.” The experimenter then covered the cup and asked the child, “How many lemons are in the cup?” as an attention check. If the child answered incorrectly, the experimenter repeated themselves once again. Once the child answered correctly, the experimenter then added one more lemon to the cup while drawing the child’s attention, “Look!” The child was then asked to indicate how many lemons were in the cup: “How many lemons are in the cup? Are there $N + 1$ or $N + 2$ lemons in the cup?” counterbalanced such that the first number word was not always the correct answer. The task therefore measured whether children know that adding one item to a set entails counting up one word in the count list. Children were tested on the following number words: 4, 7, 9, 25, 27, 29. Each number word was tested three times, such that no number word was queried again before all number words were queried at least once.

4.2. Results

As in Experiment 1, model comparisons were performed to test the effects of independent variables using likelihood ratio tests, and best-fitting models were selected on the basis of a significant chi-squared statistic and reduced AIC value. All pairwise comparisons were adjusted for multiple comparisons using a Bonferroni correction.

To reproduce the findings of Experiment 1, we compared children’s performance on the Invisible and Visible Later-Greater tasks. To do so, we constructed a base model [Accuracy (0/1) $\sim$ Task (Invisible Later-Greater/Visible Later-Greater) + (1|PID)] exploring the baseline

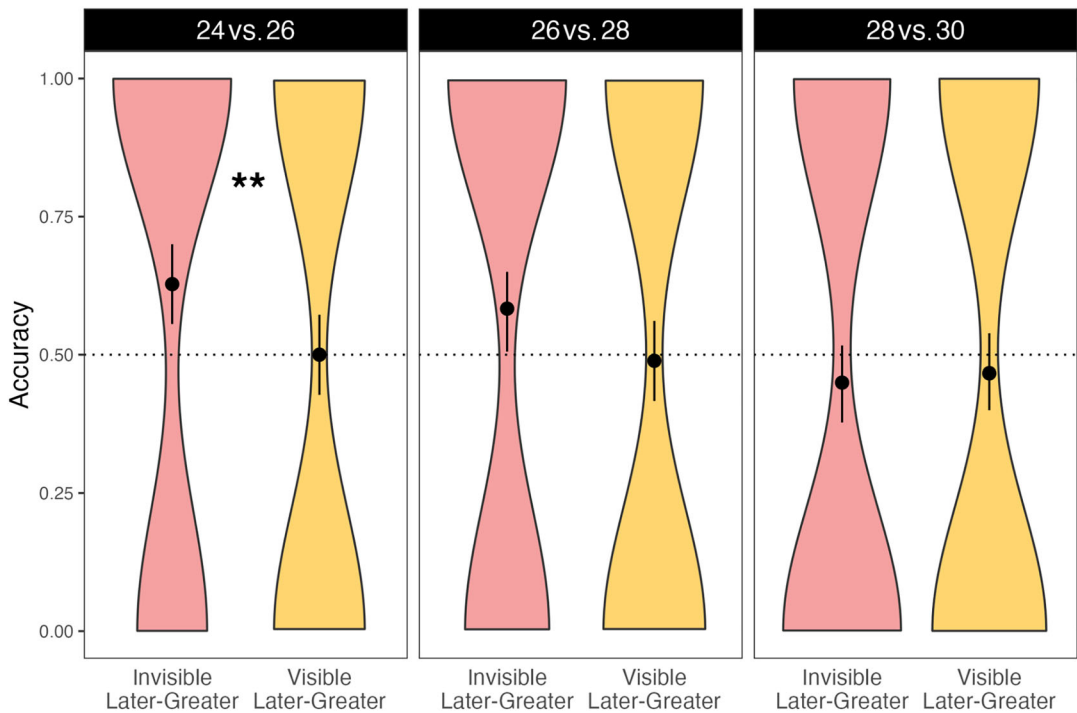


Fig. 4. Mean accuracy on numerical judgments across tasks. Each facet represents the queried comparison, and violins represent the distribution of responses for each task on that trial. The horizontal dotted line represents chance. Error bars represent 95% bootstrapped confidence intervals. Asterisks represent significant differences between tasks.

differences in performance in the two tasks. In this base model, we found that children's performance was significantly better on the Invisible Later-Greater task relative to the Visible Later-Greater task ($B = -0.28$, $p = .02$), suggesting that children used a principle to make “more” judgments. Adding a Task $\times$ Trial size interaction term to the base model improved fit ($\chi^2(4) = 13.36$, $p = .009$), reflecting the fact that children were significantly better at making “more” judgments in the Invisible Later-Greater task relative to the Visible Later-Greater task on the 24 versus 26 trial ($B = 0.54$, $p = .01$) but not the 26 versus 28 trial ($B = 0.39$, n.s.) or the 28 versus 30 trial ($B = -0.06$, n.s.), for which performance was generally poor (Fig. 4).⁶ Although adding a term for age did not significantly improve model fit ($\chi^2(1) = 0.41$, n.s.), a maximal model that included a fixed term highest count significantly improved model fit ($\chi^2(1) = 3.99$, $p = .04$) relative to simpler models, suggesting that children's rote counting ability may play some role in their “more” judgments for large numbers (Table 2).

For the Invisible Later-Greater task, children performed above chance levels on 24 versus 26 ($t = 3.53$, $p < .001$) and 26 versus 28 ($t = 2.26$, $p = 0.02$) but not on 28 versus 30 ($t = -1.34$, n.s.). By contrast, children's performance did not differ from chance on any Visible Later-Greater comparison (24 vs. 26 [$t = 0$, n.s.]; 26 vs. 28 ($t = -0.29$, n.s.); 28 vs. 30 ($t = -0.89$, n.s.)). In exploratory analyses, we sought to determine the age at which children's

Table 2
Regression models predicting children's accuracy on the three numerical comparison tasks in Experiment 2

#	Model	Coefficient Estimates (<i>B</i>)			Trial: 26v26	Trial: 28v30	VLG *	VLG *	VLG *	Summary Statistics	
		Age	HC	VLG						AIC	Log-lik
1	Task *			−0.28*				26v28		1488.3	−741.1
2	Task *			−0.54*						1482.9	−734.45
3	Trial										
3	Age + Task *	0.05		−0.54*						1484.5	−734.24
4	Trial										
4	HC + Task *		0.15*	−0.54*						1480.9	−732.5
4	* Trial										
4	Age + HC + Task *	−0.006	0.16	−0.54*						1482.9	−732.45
	Trial										

Note. Reference Group—Task: Invisible Later-Greater. Trial: 24v26. Note: A random effect of participant (1|PID) was included in all models
Abbreviations: HC = Highest Count; VLG: Visible Later-Greater.
Significance codes: ****p* < .001; ***p* < .01; **p* < .05.

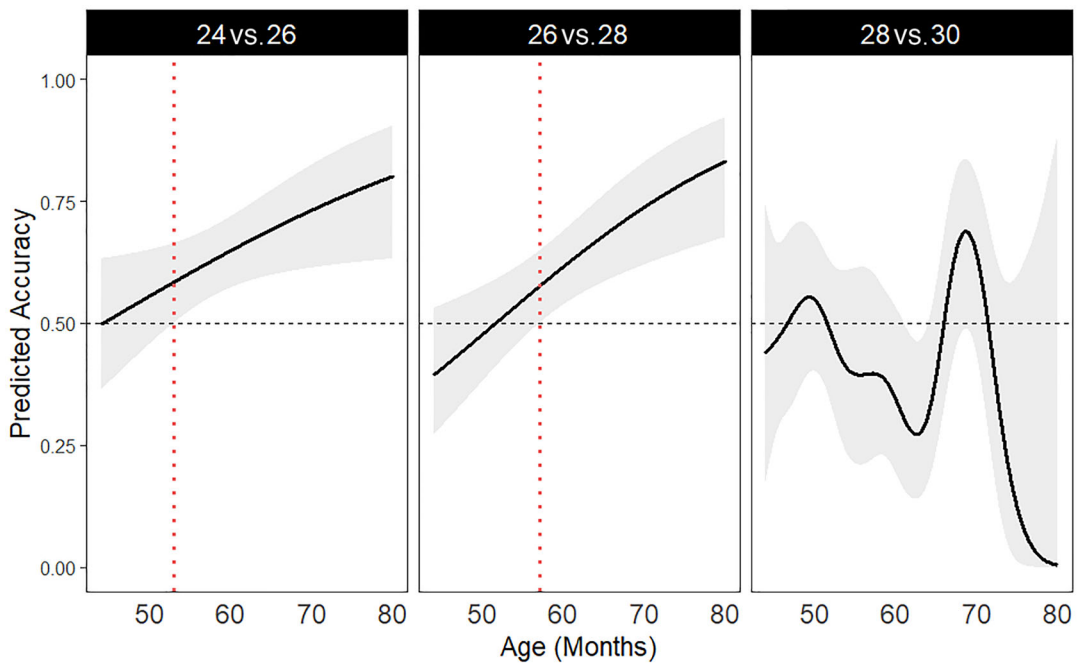


Fig. 5. Generalized additive model predictions of children's accuracy on the Invisible Later-Greater task for each trial comparison. The solid black lines represent predicted accuracy by age, and shaded areas show the 95% confidence intervals. The horizontal, dashed black line indicates chance performance (0.5), and the vertical, dotted red lines mark the estimated ages at which children's performance on each trial reliably exceeds chance.

performance on each trial of the Invisible Later-Greater task exceeded chance. To do so, we fit separate generalized additive models for each pairwise comparison of age. In each model, accuracy (1/0) was modeled as a smooth function of age in months, to capture potentially non-linear changes in performance with increasing age. For each comparison, we computed 95% confidence intervals around the predicted probabilities and identified the earliest age at which the lower bound of the 95% confidence interval exceeded chance level (0.5), indicating a shift in performance above chance. We found that children first demonstrated above-chance performance on 24 versus 26 at approximately 52.94 months and on 26 versus 28 at approximately 57.07 months. However, at no point in the tested range did performance on 28 versus 30 exceed chance (lower bound remained below 0.5; see Fig. 5). These analyses suggest that children begin to make later-greater judgments for large numbers with tight ratios around $4\frac{1}{2}$ to 5 years of age but that this ability is limited to numbers that contain unit labels between “one” and “nine.” Thus, we replicated the results of Experiment 1 but also found that children’s “more” judgments may be based on unit labels. For example, one reason children might have successfully compared 24 versus 26 but not 26 versus 28 could be that they rely on their knowledge of 4 versus 6 but struggle with relatively larger numbers like 6 versus 8.

In our next analyses, we began to explore the mechanisms by which children might acquire later-greater knowledge. First, we tested whether children might acquire the principle by an ANS induction by asking whether success on the Invisible Later-Greater task was predicted

by knowledge of mappings to the ANS. In particular, we asked whether individual differences in children's performance on the Point-to-X task—our measure of mappings to the ANS—predicted performance on Invisible Later-Greater. To do so, we constructed a generalized logistic mixed effects model predicting accuracy (1/0) on the Invisible Later-Greater task from overall Point to X performance ($\text{Accuracy} \sim \text{Age} + \text{Point to X} + \text{Trial} + [1|\text{PID}]$) and found that children's Point to X scores did not predict their “more” judgments ($B = 0.01$, n.s.). Second, we predicted children's “more” judgments on a particular trial (e.g., 24 vs. 26) from their Point to X scores on the corresponding small number trial (e.g., mappings for 6). Children were classified as either mappers or non-mappers for “six” and “eight.” To be classified as a 6-mapper, a child had to accurately identify a set of “six” on at least 4 out of 6 Point to X trials on which it was queried, including 3 trials of 24 versus 26 and 3 trials of 26 versus 28. We found no significant differences between 6-mappers and 6-non-mappers on either the 24 versus 26 trial ($B = 0.09$, n.s.) or the 26 versus 28 trial ($B = 0.29$, n.s.). Similarly, to be classified as an 8-mapper, a child had to accurately identify a set of “eight” on at least 4 out of 6 Point to X trials on which it was queried, including 3 trials of 26 versus 28 and 3 trials of 28 versus 30. Once again, we found no significant differences between 8-mappers and 8-non-mappers on the 26 versus 28 trial ($B = 0.29$, n.s.) and the 28 versus 30 trial ($B = -0.56$, n.s.; Fig. 6; Supporting Information Table S3). Altogether, these data suggest that associative mappings for unit labels do not predict children's later-greater judgments. Also, these data rule out the possibility that children's later-greater judgments for large numbers, like 24 versus 26, are based on restricting attention to mappings of just the small numbers implicated in these comparisons—for example, by reasoning that since “six” denotes a greater magnitude than “four,” then “twenty-six” must be more than “twenty-four.”

We next assessed the alternative possibility that the later-greater principle is acquired by noticing local successor relations between individual numbers. To probe this idea, we tested whether children's performance on the Unit Task, which tested knowledge successor relations between number words, predicted performance on the Invisible Later-Greater task. We constructed a generalized logistic mixed-effects model predicting accuracy (1/0) on the Invisible Later-Greater task from overall Unit Task scores ($\text{Accuracy} \sim \text{Age} + \text{Unit} + \text{Trial} + [1|\text{PID}]$). Children's performance on the Unit Task was a significant predictor of later-greater judgments ($B = 0.08$, $p = .003$). We compared this to a model that also included a term for overall Point to X performance ($\text{Accuracy} \sim \text{Age} + \text{Point to X} + \text{Unit} + \text{Trial} + [1|\text{PID}]$) and found that this did not improve model fit ($\chi^2(1) = 0.06$, n.s.). Thus, knowledge of mappings for small number words did not explain any additional variance in children's later-greater performance above and beyond knowledge of successor relations. Building on these findings in additional models, we also found that children's Unit Task performance for large numbers (i.e., naming the successors of 25, 27, 29) predicted later-greater judgments on the Invisible Later-Greater task ($B = 0.10$, $p = .02$), when controlling for age. Similarly, Unit Task performance on small numbers alone (i.e., naming the successors of 4, 7, 9) also predicted judgments on the Invisible Later-Greater task ($B = 0.13$, $p = .01$), while controlling for age. Taken together, these data suggest that children's understanding of successor relations between number words, rather than mappings between number words and the ANS, support children's inferences that later numbers represent greater magnitudes.

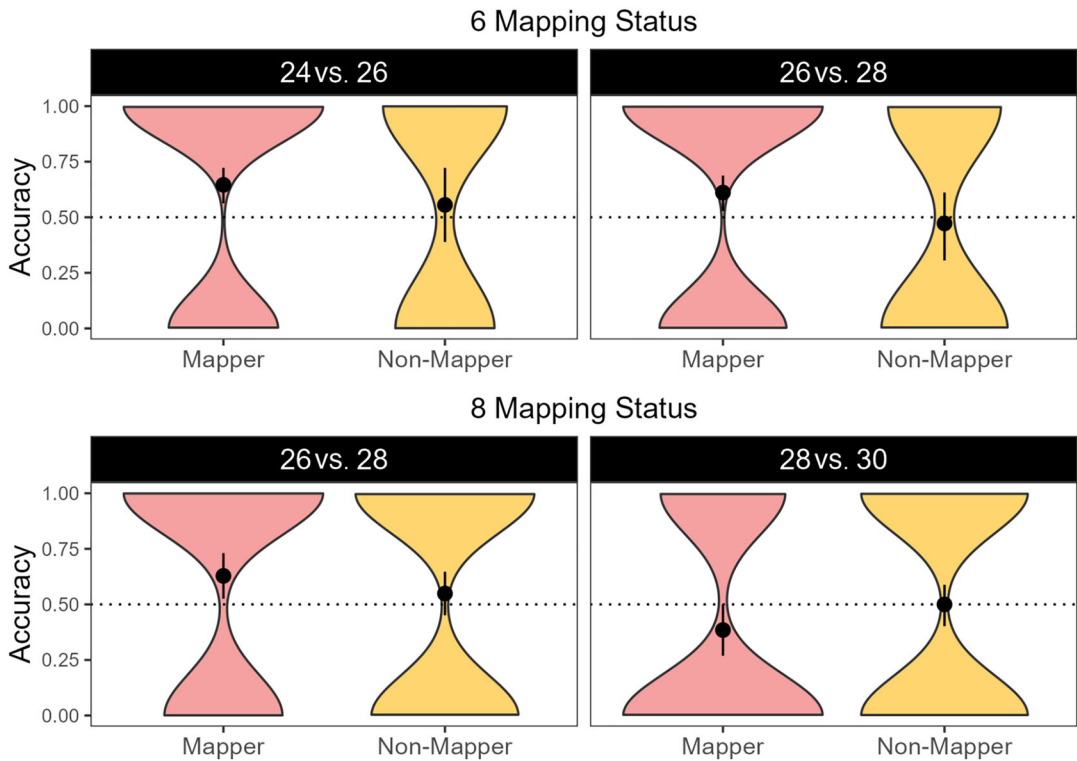


Fig. 6. Mean accuracy on Invisible Later-Greater judgments across mappers and non-mappers. Each facet represents the queried comparisons, and violins represent the distribution of responses for each task on that trial. The horizontal dotted line represents chance. Error bars represent 95% bootstrapped confidence intervals. No significant differences were found between mappers and non-mappers.

4.3. Discussion

In Experiment 2, we found two main results. First, compatible with Experiment 1, we found that children were better at making “more” judgments on the Invisible Later-Greater task than the Visible Later-Greater task. Although overall this difference was driven by the 24 versus 26 comparison, exploratory analyses found that children succeeded at 24 versus 26 by around 53 months of age and at 26 versus 28 by around 57 months of age, while children struggled with 28 versus 30 at all ages, suggesting that later-greater knowledge may first emerge at around $4\frac{1}{2}$ to 5 years of age. Second, we found that children’s “more” judgments in the Invisible Later-Greater task were significantly related to knowledge of successor relations in the Unit Task but not to knowledge of mappings of numbers to the ANS. This last result suggests that knowledge of successor relations may be a better predictor of later-greater knowledge than mappings of numbers to the ANS.

5. General discussion

We explored the origin of the later-greater principle and contrasted two ways in which this knowledge might come about—that is, via knowledge of successor relations or through mappings of numbers to the ANSs. We found two main results. First, across Experiments 1 and 2, we found evidence for knowledge of a later-greater principle. Children performed better on the Invisible Later-Greater task, which does not strictly require the ability to discriminate magnitudes, than on the Visible Later-Greater task and Point to X, which do. Also, children who had mapped number words to approximate magnitudes in the Point to X task (i.e., mappers) did not differ from children without mappings (non-mappers) when making “more” judgments in the Invisible Later-Greater task (Experiments 1 and 2). Exploratory analyses found that knowledge of the later-greater principle likely emerges around $4\frac{1}{2}$ to 5 years of age, at which point children succeeded at large number comparisons (24 vs. 26 and 26 vs. 28), though they continued to struggle with comparisons involving “thirty” even at $6\frac{1}{2}$ years of age. Second, in Experiment 2, we found that although children’s ability to map number words to magnitudes did not predict their judgments on the Invisible Later-Greater task, their knowledge of successor relations between number words (as measured by the Unit Task) did.

The current study builds on the findings of Davidson et al. (2012) and Le Corre (2014), while also presenting new evidence regarding the role of associative mappings in acquiring a generalized later-greater principle. In particular, while those studies showed that children made “more” judgments for words that had mappings to the ANS, they did not identify when children acquire adult-like knowledge of a generalized principle that allows them to make “more” judgments for non-discriminable ratios. Further, those studies did not identify how children might acquire such a principle and whether mappings to the ANS play a role. Our data suggest that although children likely make “more” judgments based on mappings to the ANS early in development, by around $4\frac{1}{2}$ to 5 years of age, they acquire knowledge of a later-greater principle that allows them to make “more” judgments independent of these mappings. Knowledge of this principle explains children’s ability to make “more” judgments for items with non-discriminable ratios, like 24 versus 26, as shown by better performance on the Invisible Later-Greater task, which does not require mappings to the ANS, relative to the Visible Later-Greater task, which does.

Our data also suggest that the way children arrive at this principle is likely independent of ANS mappings. First, we found that overall performance on the Point to X task, which measured associations between number words and the ANS, did not predict later-greater knowledge as measured by the Invisible Later-Greater task. Second, we found no difference in later-greater judgments between “mappers” and “non-mappers” in our study, unlike Le Corre and Davidson et al. This difference from past studies is likely because we used a measure of “mapper-hood” that more narrowly targeted mappings for specific number words (rather than a broad measure of children’s estimation abilities, used in past studies). In particular, we identified whether children showed evidence of mappings for numbers that were involved in specific “more” comparisons, whereas previous studies asked whether children showed a positive slope in dot-array estimates for numbers between 6 and 10 (e.g., Davidson et al., 2012; Le Corre & Carey, 2007; Le Corre, 2014). Thus, while past studies show that children with

better overall estimation abilities also have more accurate overall later-greater judgments, our study suggests that for any particular numerical comparison, there is no evidence that mappings between number words and ANS representations play a role and that instead children without mappings perform just as well as children who have them. Together, these two novel results suggest that associative mappings to the ANS may play a limited role in children's later-greater judgments.

While our results fail to support a role for ANS mappings in children's acquisition of the later-greater principle, they suggest that knowledge of successor relations between number words may play a role. In particular, children who performed well in the Unit Task, and who could judge that adding 1 to a set n resulted in $n + 1$, were more likely to make accurate later-greater judgments in the Invisible Later-Greater task. This suggests that as children learn to count, they notice that counting up one word in the count list always corresponds to adding 1 item and that therefore number words that come later in the list must represent larger quantities. Crucially, such knowledge does not require mapping individual numbers to specific approximate magnitudes, but only the knowledge that, wherever you are in the count list, counting up will always add to a cardinality. Compatible with this, while children's Unit Task performance predicted later-greater judgments, their knowledge of mappings to the ANS, as measured by Point to X, explained no additional variance.

One perhaps puzzling finding in our study is that although children made accurate "more" judgments for large number comparisons like 24 versus 26 and 26 versus 28 before the age of 5, they struggled with comparisons involving the decade term 30 and did not succeed at even $6\frac{1}{2}$ years of age. One possible explanation of this finding, which we explored in Experiment 2, is that children might perform better on items that include unit terms between 1 and 9 by simply ignoring the decade terms involved. In Experiment 2, we ruled out one version of this hypothesis, on which children use mappings between unit terms and the ANS to judge that "twenty-six" is more than "twenty-four" because "six" denotes a larger approximate magnitude than "four." Against this, we found that knowledge of mappings of small numbers to the ANS did not predict "more" judgments for larger numbers. However, another version of this hypothesis remains possible. In particular, children may know that "twenty-six" must be more than "twenty-four" based on their knowledge of successor relations between small numbers and that "six" comes after "four" in the count list. However, other explanations also exist. For example, another possibility is that some children have a generalized later-greater principle but lack familiarity with decade labels and how they are ordered with respect to other numbers outside of the rote count routine. Compatible with this, previous studies find that knowledge of decade terms is weaker than knowledge of other number words and may be more difficult to access outside of the counting routine. First, when children are asked to count as high as they can, they often stop at decade transitions, ending their counts at numbers like "twenty-nine" and "thirty-nine," suggesting less familiarity with the position of decade labels in the count list (Chu et al., 2020; Gould, 2017; Fuson 1991; Fuson et al., 1982; Siegler & Robinson, 1982; Wright, 1994). Second, and compatible with this, decade terms like "thirty" and "forty" occur with relatively low frequency in child-directed speech (Rule, Dechter, & Tenenbaum, 2015; Willits, Jones, & Landy, 2016). Finally, previous studies find that children are slow to understand how decade labels encode quantity and that each decade encodes a

multiple of 10 (e.g., Cheung & Ansari, 2023; Guerrero & Park, 2020; Miura, Kim, Chang, & Okamoto, 1988, 1994, 1999; Miura & Okamoto, 2013; Mix, Bower, Hancock, Yuan, & Smith, 2022; Mix, Bower, Hancock, Crespo, & Smith, 2024).

The present study had a number of limitations that should be addressed in future research. First, possibly complicating comparison to past studies (e.g., Davidson et al., 2012; Le Corre & Carey, 2007; Le Corre, 2014), we used the Point to X task to measure mappings to magnitudes instead of “Fast Cards” (a dot array estimation task). Here, we preferred the Point to X task because unlike Fast Cards, it tests mappings to magnitudes but does not require the child to spontaneously access and produce number words. Instead, in the Point to X task, children are given the number words so that those who know the number words can map them to visual sets even if they are not able to spontaneously produce them in a speeded task. Also, using this method, we were able to target mappings for specific number words, rather than relying on estimation slope as a measure of mapping knowledge, providing a more precise measure of knowledge for individual words. A second possible limitation of the present study is that the stimuli used in the Point to X and Visible Later-Greater tasks were not controlled for non-numerical visual features like total surface area and convex hull (for discussion, see Sella, Berteletti, Lucangeli, & Zorzi, 2017). That said, access to these numerical correlates did not appear to help children resolve trials with tighter ratios. When children succeeded at the Invisible Later-Greater task for tight ratios, they nevertheless failed in the Visible Later-Greater task, demonstrating that the quantities were not discriminable, and thus that access to these additional cues could not explain success on tight ratio trials. A final note is that recent studies have disputed whether the Give-a-Number task, as used here, provides a reliable test of cardinal principle knowledge (Krajcsi & Reynvoet, 2024). In particular, these studies contend that many children can give numbers up to, for example, 6 or 7, but not higher, drawing into question whether they have knowledge of counting principles or are subset knowers with high knower levels. Mitigating this concern, this work also finds that when such children are explicitly asked to count, they demonstrate competence with larger numbers compatible with knowledge of counting principles (Krajcsi, 2021). Also, most importantly, our method of assessing knower-level knowledge resembled that used in other studies that address the origin of later greater knowledge (e.g., Le Corre, 2014; Davidson et al., 2012), and none of our findings regarding the role of mappings in later-greater knowledge hinge on knower-level assignments.

In summary, the present study found that children do not systematically rely on item-based associative mappings between number words and the ANS to make later-greater judgments. In other words, we do not rely on this evolutionary-ancient system to judge, for example, that a salary of \$72,000 is more than a salary of \$71,000. Neither mappings for small number words nor large ones are sufficient for furnishing an understanding of a later-greater principle. Instead, knowledge of this principle might arise from noticing the successor relations between familiar number words in the count list and then generalizing this knowledge to all possible numbers, such that numbers later in the count list are always greater in magnitude.

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Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Preregistrations, data, and code are available at the Open Science Framework.

Notes

- 1 Le Corre (2014) uses the term “minimal mappers” to describe children who have associative mappings between number words and ANS representations.
- 2 The sample size was determined with an a priori power analysis on G*Power version 3.1.9.7 (Faul et al., 2009), adequate to achieve 80% power for detecting an effect of $f = 0.15$ at $p = .05$ between the three numerical comparison tasks. This power analysis was conducted to detect baseline within-subjects differences in performance between the tasks and did not account for additional variables included in the reported models. Data collection concluded once the pre-registered sample size, post exclusions, was reached.
- 3 While our adoption of this convention allows comparison to past studies, some past work has argued that there might be children who succeed at larger numbers (e.g., “six”) without having an understanding of counting, or using counting to derive their responses (e.g., Krajcsi and Reynvoet, 2024). Studies currently underway in our lab are exploring this idea.
- 4 To examine whether familiarity with 30 explained children’s poor performance on the 26 versus 30 trial of the Invisible Later-Greater task, we classified children into two groups, those with highest counts from 10 to 29 ($n = 37$) and those with highest counts of 30 or more ($n = 23$) and constructed a model predicting performance on the 26 versus 30 trial with the following formula: $\text{Accuracy} \sim \text{Age} + \text{HC_Status (Under/Over)} + (1|\text{PID})$ but found no effect of HC_Status ($B = -0.88$, n.s.). This indicates that mere exposure to number words beyond 30 did not help children make more judgments. For further exploratory analyses, see the Supporting Information.
- 5 We pre-registered that children who are able to respond correctly on five out of six comparisons on Point to X will be defined as mappers for the given trial, but felt that this

criteria may be too stringent, and may underestimate children's associative mappings to the ANS (A breakdown of children classified as mappers and non-mappers is in Supporting Information Table S2.)

- 6 Once again, we classified children into two groups, those with highest counts from 1 to 29 and those with highest counts of 30 or more. We then constructed a model predicting performance on the 28 versus 30 trial with the following formula: $\text{Accuracy} \sim \text{Age} + \text{HC_Status (Under/Over)} + (1|\text{PID})$ but found no effect of HC_Status ($B = -0.22$, n.s.).

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